

PROFILE

AI and Machine Learning Engineer with experience turning complex problems into working systems across deep learning, NLP, computer vision, XAI, and edge AI. Backed by professional R&D experience and academic research recognized at an international AI conference. Highly adaptable, proactive, and committed to building reliable and impactful AI systems.

WORK EXPERIENCE

Machine Learning Engineer - Intern
Treinetic (Pvt) Ltd | Battaramulla, Sri Lanka

June 2024 - May 2025 | 1-Year Program

- Built a facial recognition system using DeepFace + MTCNN with embedding-based identification, integrated via a Django REST API for employee attendance
- Developed TIMO - a regression-based employee working-hours prediction system using XGBoost with synthetic dataset generation and prediction APIs
- Designed an NLP-based service letter generation system with template mapping and multi-format export (PDF, DOCX, TXT)
- Created a GPT-based HRM chatbot with role-based structured prompts for user and admin access
- Collaborated across ML and backend teams; maintained version control and participated in full-cycle development from research to deployment

Stack: Python | Django | XGBoost | ARIMA | LSTM | HuggingFace | DeepFace | MTCNN | OpenAI GPT | MySQL | REST APIs

SKILLS

Programming Languages:
Python (Advanced), Java (Intermediate), SQL

Core AI and ML:
Supervised Learning, Unsupervised Learning, Model Benchmarking, Hyperparameter Tuning

Deep Learning:
PyTorch, TensorFlow, Keras, CNNs, LSTMs, Vision Transformers (ViT)

MLOps and Deployment:
Gradio, Django REST APIs, Git, GitHub, Kaggle, Docker (basic)

Computer Vision:
OpenCV, Object Detection, Object Tracking, Behavior Analysis, Image Preprocessing

Natural Language Processing (NLP):
Topic Modelling, Named Entity Recognition, Text Classification, Tokenization, Prompt Engineering, Few-Shot Learning, Dual-Prompt Strategy

Edge AI and IoT:
Raspberry Pi, MediaPipe, Embedded ML, Embedded Systems

Databases:
MySQL, MongoDB, SQLite

ACADEMIC PROJECTS

CheXplain - Dual-Level Explainable AI for Chest X-Ray Diagnostics

2026

BSc Final Year Thesis | Paper accepted at IVP AI 2026

- Chest X-rays are meaningless to patients and risky when misread by doctors. I built a system that doesn't just classify pathologies, it explains *where* and *why* in plain language, for both clinicians and patients. The core challenge: making Grad-CAM heatmaps actually useful as LLM context. Solved it with a novel Behavior Extractor module that maps activation zones to anatomical regions. Benchmarked across 4 vision architectures and 5 LLMs to find what actually works at scale.

Stack: PyTorch | DenseNet-121 | ViT · Grad-CAM | HuggingFace | Qwen2.5 | LLaMA 3 | Gradio | scikit-learn

Public Opinion NLP Pipeline - Education Policy Analysis

2026

Natural Language Processing | BSc Module Project

- Applied TF-IDF, K-Means clustering, and LDA topic modelling across 11 education themes; benchmarked classical ML vs. fine-tuned transformers (BERT, DistilBERT, GPT-2).

Stack: NLTK | spaCy | scikit-learn | HuggingFace | BERT | DistilBERT | Mistral 7B | Qwen 2.5 | VADER | LDA | TF-IDF

Battery Storage Optimization - Hybrid LSTM + Genetic Algorithm Framework

2026

Computational Intelligence | BSc Module Project

- Implemented NSGA-II (multi-objective: cost, peak demand, battery degradation) producing 12-14 Pareto-optimal solutions; peak demand reduced by 17.1%

Stack: Python | PyTorch | NSGA-II | LSTM | NumPy | Pandas | Matplotlib | scikit-learn

WakeVision - Edge AI Driver Drowsiness Detection System

2025

Edge AI & IoT Systems

- Most drowsiness detection systems need a phone or cloud connection. This one run entirely on a Raspberry Pi 4, no internet, no latency, no excuses. Multi-task model handles eye closure, yawning, and gesture recognition

Stack: Python | PyTorch | MediaPipe | OpenCV | React Native | MongoDB | Raspberry Pi OS

FishFit - Fish Health Behavior Detection System

2024

Data Science Group Project | Haxmas Hackathon Champions - Ascentic Pvt Ltd

- Built real-time object detection + tracking pipeline to analyze fish swimming patterns and classify health status (normal / abnormal) using motion and trajectory features

Stack: Python | Django | MySQL | OpenCV | JavaScript | HTML/CSS

EDUCATION

BSc (Hons) in Artificial Intelligence and Data Science - First Class Honors

January 2023 - January 2026

Informatics Institute of Technology (IIT) · Robert Gordon University (RGU), Aberdeen

Foundation in Higher Education - Distinction

May 2022 - December 2022

Informatics Institute of Technology (IIT)

G.C.E. Ordinary Level - 8A Passes

May 2022

Isipathana College

CERTIFICATIONS & AWARDS

- **Haxmas Hackathon Champion** - Ascentic Pvt Ltd: Won island-wide AI competition for FishFit (fish health behavior detection using CV & object tracking), 2024
- **Python for Beginners** - University of Moratuwa: Completed Python fundamentals with real-world problem-solving applications
- **IVPAI 2026 Paper Acceptance** - Paper accepted: "Explainable Chest X-Ray Diagnosis Using Visual Saliency and LLM-Generated Dual-Audience Clinical Narratives," International Visual Intelligence & Perception in AI Conference, 2026

REFERENCES

Mr. Banu Athuraliya

Module Leader | Visiting Lecturer |

Final Year Project Supervisor

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